

A Taxonomy for Design Requirements from Corporate Customers

John K. Gershenson¹ and Larry A. Stauffer²

¹Mechanical and Aerospace Engineering, Utah State University Logan, Utah; ²College of Engineering – Boise, University of Idaho, Boise, Idaho, USA

Abstract. *We have developed a taxonomy that classifies those needs of a corporation that impact product design. We call these needs corporate requirements. In contrast to the consumer or end-user requirements, corporate requirements come from internal sources such as marketing, finance, manufacturing, and service. This taxonomy allows for an organized method of gathering, managing, and retrieving the requirements. The taxonomy also helps to facilitate a broader, clearer form of Quality Function Deployment. Generic in nature, this taxonomy provides a template with which to create taxonomies for a given product within a given company or industry. We include an industrial case study to demonstrate this concept.*

Keywords: Design; QFD; Requirements; Taxonomy

1. Introduction

The development of the taxonomy described in this paper is an outgrowth of research into customer-focused design. The development is based upon previous definitions of taxonomies in many different fields. Using these definitions, we were able to prescribe a method for developing taxonomies, and apply it to the creation of a taxonomy for corporate requirements.

1.1. Motivation

The inclusion of customer desires is integral to the design of quality products (Ramaswamy and Ulrich 1992). Several methodologies help companies include these desires in their products. The most popular methodology is Quality Function Deployment (QFD), graphically represented by the House of Quality (HOQ) (Hauser and Clausing 1988). HOQ provides a

framework for listing customer requirements, design attributes, trade-offs, and other information focused on meeting customer needs.

In QFD, customer requirements typically refer only to consumer or end-user requirements. End-user requirements are expectations that the end-user has for the product, often called the ‘voice of the customer’. Marketing groups, which gather and analyze end-user information, often control the initial product concepts in this process. However, it is necessary to consider all customers, not just end-users, from the beginning of the design process, and to give all members of the design team a voice, thus increasing the value of the product. Managing the requirements early in the design process allows them to be more thoroughly considered. This practice saves time and money in the end, since it reduces expensive downstream changes.

1.2. Taxonomies

Taxonomies, like the one discussed in this paper, are commonly used to classify large bodies of information. Classification adds order and clarity to information (Derr 1973). ‘Classification is the assignment of like objects to recognizable groups. It facilitates reference to them and the transmission of information about them’ (Jeffrey 1982). It is important to note, though, that it is never possible to classify a whole subject; it is only possible to classify our knowledge of it. This limitation implies that our knowledge of any subject is neither exhaustive nor objective.

Taxonomies, formally or informally, are used frequently in engineering design as a way to organize information. Computer Aided Engineering (CAE) applications, which are menu driven, have choices which constitute a taxonomy. When engineers click through pull down menus to apply shapes, shades, etc. they are searching through a taxonomy. Within the field of mechanical design, one well known taxonomy

Correspondence and offprint requests to: J. K. Gershenson, Mechanical and Aerospace Engineering, Utah State University, Logan, UT 84322, USA.

is Ullman's taxonomy of mechanical design (Ullman 1989). Ullman breaks down all design problems into solvable chunks and characterizes (i.e., classifies) the design process. The resulting taxonomy classifies design problems and tools to solve the problems in terms of environment, problem, process and research approach. However, this taxonomy does not classify product requirements. Stauffer and Slaughterbeck-Hyde (1989) discuss a taxonomy that includes lifecycle design considerations. They suggest the use of such a taxonomy for product requirements, but do not develop it. VDI 221 presents a taxonomy for developing design requirements.

1.2.1. Taxonomy Characteristics

Taxonomies are common to many fields of science. The rules for creating taxonomy classifications have two common threads. Derr (1973) states that classes must be mutually exclusive and exhaustive. Dunn and Everett (1982) state that a '[t]axon [or a class] is a taxonomic of any rank that is sufficiently distinct to be worthy of being assigned to a definite category'. This leaves the classification to be subjective. In addition, Slaughterbeck-Hyde (1989) discusses depth or parallel structure as an important consideration in developing a taxonomy. This yields three, interrelated issues that characterize a taxonomy: completeness, perceptual orthogonality, and parallel structure. Each issue is necessary to have a well constructed, useful taxonomy.

- *Completeness*: a taxonomy should be 'exhaustive over a domain' (Slaughterbeck-Hyde 1989). It should be receptive to any element of its domain. Completeness is impossible to obtain but it is useful as a goal. It is important that a taxonomy include all information necessary to make a quality product. However, storing excess information is harmful to the design process, in that it forces designers to consider information which does not yet have a bearing on their efforts, and does not yet translate into design attributes; resulting in wasted elicitation and design time. Therefore, all 'possible' information must be trimmed to all 'useful' information. This trimming from the multitude of possible corporate issues is justified by the 80:20 heuristic or Pareto's law of distribution of costs (Dieter 1983). Pareto's law states that the top 20% most important corporate issues generally represent about 80% of a product's problems. Although the ratio may change depending upon the product or class of problems, in general, a small number of

corporate issues, where problems occur most frequently, will have a large impact on product quality.

- *Perceptual Orthogonality*: a taxonomy is perceptually orthogonal when each classification or taxon is perceived as being mutually exclusive of all others. Therefore, no piece of information should be perceived to be classified under two categories. 'For example, in a taxonomy of trees, one might have two categories; 1) deciduous and 2) conifers. This taxonomy would be orthogonal in that a tree can be entered under one category or the other, but not both' (Slaughterbeck-Hyde 1989). Orthogonality is important in this taxonomy to maintain distinctness between the taxons. If the taxons are not distinct, it will be unclear where to store or retrieve requirements, and the taxonomy will no longer aid the development process.

The judgment of whether a taxonomy or a group of taxons is perceptually orthogonal is subjective and domain dependent. For instance, there may be a tree that is both deciduous and a conifer, but Slaughterbeck-Hyde did not know of one and therefore created those taxons. There is no set of rules or methodology for developing a perceptually orthogonal taxonomy. For the taxonomy described in this paper, the common delineations existing in the literature were used as a starting point. From there, classifications were reviewed for both perceptual orthogonality and completeness. The orthogonality of the taxonomy is best exemplified by the top tier of a taxonomy discussed in Section 2.1.

- *Parallel Structure*: is the consideration of the hierarchical classification of requirements. Requirements at an upper level of the taxonomy are more broad or general than those at a lower level. It is important to make the hierarchical levels of each branch of the taxonomy uniform in their abstractness. This uniformity eases the management and understanding of the taxonomy. 'It is common knowledge that design information exists at various levels of abstraction all along the design process. However, there is no commonly agreed upon categorization or understanding of these levels' (Baya 1994). Corporate requirements, like all design information, can be of varying levels of detail or abstractness. We therefore created the taxonomy levels to insure uniform abstractness across each level of the corporate requirements taxonomy.

1.3. The Corporate Requirements Taxonomy

Our taxonomy of customer requirements is partitioned into taxonomies for four requirement types (Gershenson and Stauffer 1995): end-user, corporate, technical and regulatory requirements (Fig. 1). This breakdown is based upon the source from which requirements come, i.e. the end-user, the producer itself, society, and mother nature. End-user requirements include the end-users' expectations about the product's capabilities, aesthetics and usability. Marketers usually gather these end-user requirements. Corporate requirements encompass business issues as well as product lifecycle issues. Corporate issues are of concern to the individuals involved in the related engineering and non-engineering disciplines. These individuals are commonly the source for the requirements. Regulatory requirements include safety/health, environmental/ecological, disposal and/or political issues. Government agencies and empowered private groups often publish regulatory requirements and product standards. Technical requirements include such things as engineering principles, material properties and physical laws. Handbooks that detail some of the engineering principles behind the designs are useful resources for technical requirements. User context characterizes the end-user of the product, helping the designer apply the four types of requirements. As such, it is not one of the four requirement areas of the customer taxonomy.

We have constructed taxonomies in two of the four requirements areas. The end-user requirements taxonomy was created first (Hauge and Stauffer 1993). The corporate requirements taxonomy is presented in this paper. Regulatory requirements and technical requirements taxonomies are not being developed for two reasons. First, most designers find this information the easiest information to find.

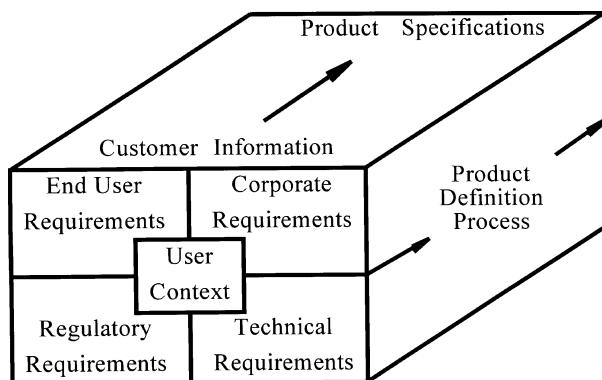


Fig. 1. Requirements cube showing the various types of requirements and how the information fits into the product definition process.

Both regulations and technical knowledge are well documented. Secondly, the information used in these two areas is more context-dependent than in corporate requirements. Any taxonomy created would be either too broad for use on any one product, or too specific for use on more than one product. We developed these taxonomies for use in a methodology for managing product definition information. We named this methodology a Methodology of Organizing Specifications in Engineering (MOOSE) (Gershenson et al. 1994; Gershenson and Stauffer 1995).

1.4. A Customer Requirements Taxonomy Application in MOOSE

MOOSE uses the taxonomies described in this paper to elicit, store and retrieve product definition information. The taxonomic structure guides these design activities, and allows for easy inheritance of knowledge into future products. MOOSE can be especially useful for novice designers who know little of where to begin in their search for product definition information. In addition, adding structure to the information helps larger organizations manage their design process. Thus, when filled with appropriate information, the taxonomies supply design teams with the knowledge necessary for defining the requirements of a product, as well as the means for storing and using the requirements.

MOOSE helps in generating product specifications and, while facilitating QFD, leads to high quality products. MOOSE also provides an active design history. Providing an active design history is an important facet of design and design recording (Chen et al. 1990). Design histories increase design productivity by increasing information reuse, decreasing reoccurring mistakes, and increasing awareness of potential problems.

A natural question at this point is. 'What are the differences between MOOSE and QFD?' QFD aims at improving three product metrics: development cost, time to market, and the customer's perception of the product's quality (King 1987). The product team first records a list of what customers desire in a product – customer requirements. Next, engineers record a series of design attributes – quantifiable parameters – that satisfy the customer requirements. A matrix keeps track of the effectiveness with which each attribute satisfies a requirement. The QFD matrix also provides spaces to benchmark one's own product against others, and a matrix showing trade-offs between attributes (Bossert 1991; Chen and Ward 1991; King 1987).

Although QFD provides the basic framework for concurrent design activities, it could be more robust. The graphical representation causes information to be cryptic, reducing the communication of design rationale except to those responsible for building it. This limits the value of QFD for redesign. Despite the open nature of QFD, the customer requirements are usually limited to end-user requirements, leading to costly redesigns later in the design process. Although QFD is applicable to any customer requirement, it gives the designer little guidance as to which customer requirements may be needed. In addition, the customer requirements are not very clear to a design engineer and may require additional explanation. Finally, QFD represents trade-offs as a positive or negative relationship, which greatly simplifies the true relationship between design requirements. The above limitations of QFD prompted the creation of MOOSE.

MOOSE encourages the inclusion of all customer requirements, together, in a single form, understandable by all facets of the organization, and ready for comparisons and trade-offs. MOOSE's taxonomy facilitates the inclusion of all pertinent customer requirements by giving the design team the guidance necessary to develop a broader set of requirements. MOOSE tackles the graphical limitations of the HOQ by allowing an in-depth description of the customer requirement and a corresponding level of criticality. MOOSE also expands the benchmarking of a product to include a fuller description of similar products and components. In addition, MOOSE represents trade-offs between customer requirements with full descriptions of all the implications on each requirement. Finally, MOOSE facilitates the redesign process by storing multiple design histories that include the rationale behind the design specifications. The taxonomy helps design teams store and retrieve this information easily and logically.

2. Corporate Requirements Taxonomy

The corporate requirements taxonomy is the classification system for product requirements which reflect the needs of the corporation regarding the product. Just as the end-user imposes various expectations regarding usability, function, etc. on a product, corporate customers (or stakeholders) impose their expectations as well. A corporation's various functional groups bring their own particular concerns which can impact on the form and function of the product. For example, manufacturing's assembly

capabilities may dictate using screws or snap fits, and concerns from shipping personnel could restrict product orientation. The taxonomy is used for gathering, storing and using requirements. Therefore, this taxonomy will act as a set of cubbyholes for the requirements, with all applicable requirements going into the appropriate cubby, labeled with a taxon. Taxons have been arranged in three levels: the *functional level*; the *task level*; and the *attribute level*.

The *functional level*, the most abstract level, contains requirement categories. Each category is an aspect or functional group of the product lifecycle, also called a 'lifecycle issue'. These issues include marketing, manufacture, service, retirement, etc. The headings for each issue are unique. In addition, level one describes each issue in a single tier of the taxonomy without using sub-tiers. This level is used in parallel with product definition.

The *task level* contains the tasks that describe what must be done to accomplish the level one lifecycle issue or function. To delineate all the individual tasks that make up each lifecycle issue, multiple tiers of the taxonomy may be necessary. Therefore, the *task level* may be made of many sub-tiers that operationally define the task. For example, 'shipping' is operationally defined as 'packaging', 'warehousing and handling', and 'transporting'. In addition, although the path or string of characterizations will be unique within each lifecycle issue, the taxonomy headings may be similar or identical to those within a different lifecycle issue. This level is used in parallel with conceptual design.

The *attribute level* contains those attributes of the product that are affected by the tasks and about which the requirements pertain. Therefore, the taxonomy at this level characterizes the *task level*. These taxonomy headings are unique within a particular task, but, as the set of product attributes is limited, each task may have similar or identical attributes. This level is used in parallel with embodiment design.

Our entire taxonomy may not be appropriate for every application. When creating a taxonomy for use with a specific product or family of products, it is necessary to create a unique taxonomy that takes into account the specific needs of the product. Usually this can be done by trimming, deleting areas of the taxonomy that do not apply. Our taxonomy covers most of the important facets of each issue where most quality problems occur (based on Pareto's law). In addition, product design is not yet complete when this taxonomy is being used, it must still go through additional phases of design where requirements will be viewed in more depth and focus. Therefore, complete coverage is not as important as eliminating

the majority of quality problems. Finally, it is important to keep in mind that the goal of the taxonomy is to organize only those requirements that affect product design.

Development of the corporate requirements taxonomy began with an exhaustive literature search for classes of product requirements. Those classes which dealt with any corporate issues (be they internal, how they deal with their end-users, or how they deal with their suppliers) were then listed as potential upper level taxons. The taxons were trimmed and shifted using the classifications found in literature, as well as the principles of completeness, perceptual orthogonality and depth discussed in the previous section. The same routine was used in the further breakdown of each of these classes. The manufacturing section, being so accountable to cost, was then developed and tested with great rigor. See Gershenson and Stauffer (1996) for more detail.

2.1. Functional Level

The functional level of the corporate requirements taxonomy is broken down into nine taxons. Each taxon represents a functional group of the corporate requirements. The functional level is shown in Table 1.

For the purposes of this taxonomy, the following definitions apply:

Marketing: any corporate driven marketing issues that affect product design, including the process by which marketing is done, but excluding all findings by marketing for end-user requirements.

Business Environment: any issues concerning the structure of the business and the world it operates in that affect product design, including personnel and competitors, but excluding strategic decision making.

Strategic Management: any issues concerning the strategic plans of the company that affect product design, including the process of making these plans, but excluding any non-strategic business issues.

Table 1. Functional level of the corporate requirements taxonomy.

1 Corporate Requirements:	
1.1 Marketing	1.6 Manufacture
1.2 Business Environment	1.7 Distribution
1.3 Strategic Management	1.8 Support/Service
1.4 Finance	1.9 Retirement
1.5 Accounting	

Finance: any issues concerning money supply of the company and the environment it operates in, excluding strategic decision making and auditing the financial status of the company, that affect product design.

Accounting: any issues concerning the auditing and record keeping of the financial status of the company that affect product design.

Manufacture: any issues concerning the production of the product that affect product design, including facilities and processes necessary to do so.

Distribution: any issue concerning the movement of the product from the end of manufacturing to the end-user that affect product design, including preparing the product for use by end-users (customer readiness) and packaging but excluding support and service.

Support/Service: any issue concerning the maintenance, repair or support of the product that affect product design, including informational requests.

Retirement: any issue concerning the end of the products useful life that affect product design, including reuse and remanufacturing among many other things, but excluding resale.

This corporate requirements taxonomy was developed by an exhaustive literature search which uncovered sources such as: Stauffer (1988), which develops a set of lifecycle design considerations including schedule, manufacture, transportation, performance, maintenance and disposal; Hauge and Stauffer (1993) and Keys (1990), which develops 'lifecycle customer requirements' that included function, features, materials, readiness, service, retirement; and which discuss the issues of the lifecycle including product/market research, manufacturing process, qualification (testing), reliability issues and customer service/maintainability/supportability and also talk of product migration through the organization functions of technical research, marketing, design, reliability, manufacturing, quality assurance, sales, distribution, service and support. Other references were used including Dieter (1983), Gershenson and Ishii (1991), Kusiak (1993), Ullman (1989), Fiksel (1991), Anderson (1990) and Vonderembse and White (1991).

One goal in arranging the functional level taxons was to arrange them to match different fundamental groups, if possible. The use of additional marketing and management texts, such as Wright et al. (1992), Hundal (1991) and Wheelan and Hunger (1989), allow for a more realistic breakdown of the 'business areas'. This breakdown better reflects the actual method of business operation and classification of

business functions. Bralla (1986) has an in-depth coverage of manufacturing issues, which was used to better understand the manufacturing functional group. Prototyping and testing/inspection were moved under the category of manufacturing where they are usually handled. In Stauffer (1988) transportation is broken down to include packaging, shipping, and storing. After looking at other texts such as Coyle (1990), we put all of these issues under the heading distribution, as this is commonly the name of the department that handles this work, and included customer readiness which they often oversee. Finally, looking at our definition of environmental compatibility and comparing it with discussions on 'Design for Environmentability' (Navinchandra, 1991) and other literature on the topic, we felt that, outside of product retirement issues, environmental concerns were normally the responsibility of the individual functions within the corporation, and not the responsibilities of a separate group. Therefore, a note was made to include environmental concerns in the taxonomies for each of the issues.

2.2. Task Level

The list of taxons shown in the previous section was then broken down into a task level depicting the tasks performed by each of the above functional taxons. We used 38 taxons to describe the primary tier of the task level, 285 for the secondary level, and six repeated taxons for the third level. The primary tier of the task level of the taxonomy is shown in Table 2. The complete, three-tiered representation of the task level of the taxonomy is not shown here because of space limitations, but is detailed in Gershenson and Stauffer (1995).

Marketing: comparing the many breakdowns of the field developed these five basic tasks of marketing. Vonderembse and White (1991) list the tasks of the marketer as they appear above. Wheelan and Hunger (1989), as most of the available literature, concern themselves with things that the marketer is concerned with product, place, promotion and price. These easily fit into the five basic tasks, with market investigation and estimation of volume taking in all those elements

Table 2. Beginning levels of the task level of the corporate requirements taxonomy

1 Marketing	
1.1 Market investigation	1.4 Develop distribution system
1.2 Estimation of volume and price	1.5 Initiate marketing
1.3 Formation of Market Strategy	
2 Business Environment	
2.1 Societal	2.3 Internal
2.2 Industrial	
3 Strategic Management	
3.1 Environment scanning	3.3 Strategy implementation
3.2 Strategy formation	3.4 Evaluation and control
4 Finance	
4.1 Description of organizational financial status	4.4 Capital formation
4.2 Financial analysis	4.5 Profit performance evaluation
4.3 Return on investment analysis	
5 Accounting	
5.1 Data for cost estimation	5.3 Feasibility of plan
5.2 Productivity analysis	5.4 Collect cost data
6 Manufacture	
6.1 Part Production	6.4 Top Level Assembly
6.2 Finishing	6.5 Inspection/Quality Assurance
6.3 Joining	
7 Shipping	
7.1 Packaging	7.3 Transportation
7.2 Warehousing and Handling	
8 Support/Service	
8.1 Diagnostics	8.3 Repair
8.2 Maintenance	8.4 'Customer Support'
9 Retirement	
9.1 Reuse	9.4 Recycle
9.2 Product Extension	9.5 Disposition
9.3 Remanufacture	

that are considered under product, developing a distribution system and a small part of formulation of marketing strategy covering the elements of place, formulation of marketing strategy and initiation of marketing cover all of promotion and estimation of price being the same as price. Trucks (1987) utilizes market research, buying and selling, transportation and storage, risk taking, standardization, and financing all as the elements of marketing. However, no other sources agree with this expanded definition of the role of marketing that seems to incorporate other lifecycle issues into it.

Business Environment: the elements of the business environment, understanding and designing for societal, industrial and internal environments, are shown as suggested by Wheelan and Hunger (1989). The only difference is that Wheelan and Hunger use the task environment in place of industrial environment. Wright et al. (1992) discuss macro environment and industrial environment as elements of business strategy and marketing, respectively. However, their definition of the business strategy has more to do with some elements of strategic management than understanding the business environment. Similarly, Wright et al.'s breakdown of the industrial environment under marketing closely parallels Wheelan and Hunger's breakdown of the industrial environment within business environment.

Strategic Management: the elements of strategic management are also shown as suggested by Wheelan and Hunger (1989). As mentioned previously, Wright et al. (1992) discuss elements of strategic management. Wright et al. break business strategy into the elements of the various environments which are scanned instead of listing the tasks the people must perform. However, when breaking down these environments, they list many of the issues involved in strategic management rather than business environment. It is our feeling that the entire lifecycle issue of strategic management may, when reviewed by experts in the field, be discounted.

Finance: finance and accounting are business lifecycle issues outlined by Vonderembse and White (1991). For finance, they list the tasks shown in our taxonomy. These elements agreed with most other texts that did not spell them out as clearly. It is our feeling that the entire taxon of finance may, when reviewed by experts in the field, be discounted due to a lack of effect on the details of product design.

Accounting: accounting went through the same development as finance with the same justifications. It is our feeling that the entire taxon of accounting may, when reviewed by experts in the field, be

discounted because although it is important to the company, it typically has no direct impact on product design.

Manufacture: the most detailed section of the taxonomy is manufacturing. The development involved an extensive literature search. Afterwards, experts in the manufacturing field screened the initial classification. Gershenson and Stauffer (1995) detail this process.

Shipping: Coyle et al (1990) discuss shipping but, in addition to the elements shown in the taxonomy, they list inventory control, purchasing, facility location, order processing, customer service, and pricing. Many marketing texts including Wheelan and Hunger (1989) and Trucks (1987) state that shipping is controlled by marketing. However, if the goal is design requirements, a shipping department will know more about packaging, warehousing and transportation than a marketing department. However, for elements such as inventory control and pricing, marketing is the department to approach. In addition, facility location and purchasing are strategy issues, while order processing and customer service, both customer service issues, are discussed below under support/service.

Support/Service: support/service is broken down into the four tasks of diagnostics, maintenance, repair and 'customer support'. The break down into these four tasks is based upon work by Gershenson and Ishii (1991) and a few automotive companies (Makino 1989; Anderson 1990; Goldman and Slattery 1964). Diagnostics is the task of discovering the cause of trouble with the product or its use. Maintenance is the task of adjusting, tuning and/or replacing aspects of the product or its use in order to maintain proper functioning. This process is done whether or not the product is functioning properly. Repair is the task of correcting a malfunctioning aspect of the product. 'Customer support' is the task of aiding end-users in using or understanding the product. Anderson (1990) suggests the inclusion of a spare parts availability within service. While this is an important production and business concern, it is not a major design concern.

Retirement: designing for product retirement is a relatively new phenomenon. Henstock (1988) details six elements of what he calls recycling: collection, separation, identification, landfill, incineration and remanufacturing. He eliminates the environmental impact on packaging, saying '[p]ackaging was specifically excluded from the study because ... it was virtually inconceivable that changes would ever be made for the sake of recyclability'. The first three

of Henstock's elements (collection, separation and identification) do fall under recycling, but landfill and incineration are clearly manners of disposition, not a reuse of raw materials, and remanufacturing is, in itself, another means of retiring the product. This, coupled with Navinchandra's (1991) discussion of product retirement, which includes reuse and product extension with recycling and disposition, led to the elements listed in the taxonomy above.

2.3. Attribute Level

The attribute level of the taxonomy is made up of those elements of a product that a designer may specify. Each attribute is an aspect of the product that must be described to have a complete product description. The attributes shown in the taxonomy are those aspects of the product that the task level taxons affect. Figure 2 details the definitions of the product attributes.

Definition of Product Attributes

Geometry: The shape and dimensions of the product.

Feature: A characteristic of a part that can be identified by some generic behavior. It has significance in design, analysis, manufacturing, or some other product domain (Shah, 1991).

Surface Condition: The condition of the outer surface of the product. This includes the color, texture, surface finish, and other surface attributes.

Tolerances: The dimensional and locational variation of the product and also the variation in mechanical properties resulting from processing operations (Shigley, 1983; Shah, 1990). This includes the source for any allowances.

Material: The materials that compose the product.

Facilities: The facilities available to process the product during any time of its life-cycle. These could include milling machines, service tools, recycling shredders, etc. It also includes the environment that the facilities create for the process.

Fig. 2. Definition of product attributes.

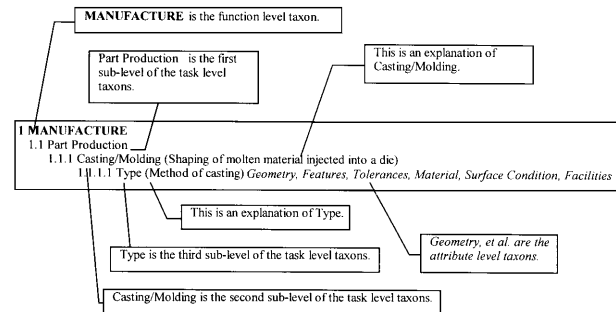


Fig. 3. A key explaining the structure of the corporate requirements taxonomy.

2.4. An Example Section of the Taxonomy

Figure 3 is a key for reading the taxonomy, and understanding which taxons belong to which levels. To show how all of the levels of the taxonomy are represented together, we present Table 3, which shows the full depth of one small piece of the manufacture taxonomy (casting and molding). Function level taxons are shown in bold and all capitals, task level taxons are shown in plain text, and attribute level taxons are shown in italics. Note that only the taxons are shown, and that at the task level the taxons range the breadth of tasks that the machine, the product, and the operator perform during the process. The taxons are used to classify the requirements; the actual requirement is stored under any of the taxons. Requirements for the product may or may not occur at any taxon level. The requirements are shown in the example of the following section. Page limitations for this paper make showing all of the taxonomy categories prohibitive. However, readers interested in seeing the entire taxonomy can refer to Gershenson and Stauffer (1995).

3. An Example of Taxonomy Application

An example was developed to demonstrate the role of the corporate requirements taxonomy in product design. The case study also served to shed light on answers to two remaining questions: how to elicit requirements; and in what form requirements should be stored. The initial stages of design (product definition and conceptual design) were completed for a shorting connector by Schwieter Engineering Labs (SEL), a manufacturer of power protection equipment. Manufacturing requirements were of primary concern during this exercise, in which we gathered customer requirements and completed product definition, fully documenting the process. We also developed concept alternatives and a

Table 3. The complete, in-depth corporate requirements taxonomy for the casting/molding task**1 MANUFACTURING****1.1 Part Production****1.1.1 Casting/Molding (Shaping of molten material injected into a die)**

- 1.1.1.1 Type (Method of casting) – *Geometry, Features, Tolerances, Material, Surface Condition, Facilities*
- 1.1.1.2 Gating (Placement of the material inputs) – *Geometry, Tolerances, Facilities*
- 1.1.1.3 Die (Mold used for casting) – *Geometry, Features, Tolerances, Material, Surface Condition, Facilities*
- 1.1.1.4 Venting (Allowing heat to escape mold) – *Geometry, Surface Condition, Facilities*
- 1.1.1.5 Distortion (Change in the shape or properties of the product due to the casting process) – *Geometry, Tolerances, Material*
- 1.1.1.6 Parting (Separation of the product from the mold) – *Geometry, Features, Surface Condition, Facilities*
- 1.1.1.7 Force (Pressure used to form the product) – *Geometry, Features, Material, Facilities*
- 1.1.1.8 Allowances (Variation in the shape during casting) – *Geometry, Material, Facilities*
- 1.1.1.9 Curing (How the product is solidified during casting) – *Geometry, Material, Surface Condition, Facilities*
- 1.1.1.10 Rate (Number of products cast/unit time) – *Geometry, Tolerances, Material, Facilities*
- 1.1.1.11 Agent (Substrate used to aid the flow, forming, or cooling of the material) – *Material*
- 1.1.1.12 Core (Additional die used in the casting) – *Features, Surface Condition, Facilities*
- 1.1.1.13 Insert (Part that is imbedded in the casting) – *Features, Surface Condition, Facilities*
- 1.1.1.14 Part Cleanup (Removal of gates, risers, etc.) – *Features, Tolerances, Material, Surface Condition, Facilities*

conceptual design of the switch. Problems that occurred during the process, as well as comments from the company about the methodology were documented. The taxonomy was used to gather, store and recall corporate requirements, especially manufacturing requirements, used in the redesign of the connector.

pair makes contact in two places. These clips hold a relay that can be pulled in and out by power company workers doing maintenance. Because it is necessary to prevent an open circuit at any time, the two contact points ensure that the clips will always be touching each other or the metal relay at one of the contact points at all times.

3.1. Example Product

The SEL shorting connector is a palm-sized, plastic casing with three pairs of metal clips (Fig. 4). Each

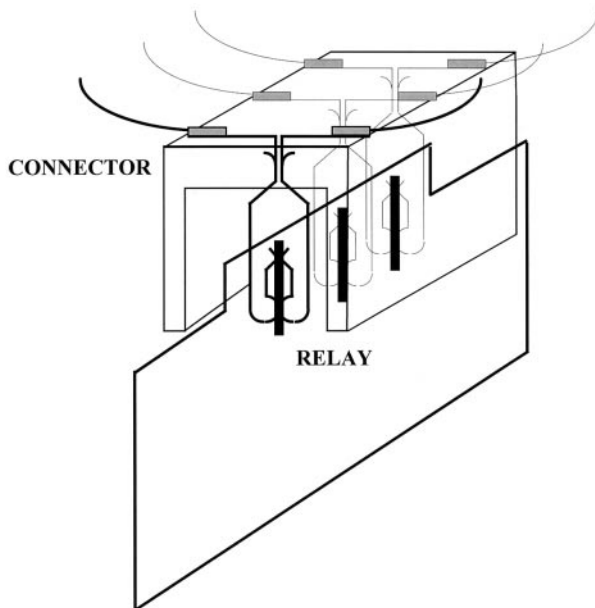


Fig. 4. The shorting connector showing the three sets of pins, each with two contact points which connect to the relay to maintain the circuit.

3.2. Eliciting Information

The first task in the example was to gather requirements to define the product. Our corporate requirements taxonomy was used as a checklist for asking questions of a product engineer from SEL. The product engineer felt that starting at higher levels of the corporate requirements taxonomy and progressing towards more detailed issues was the easiest way to proceed with elicitation. He was able to give information leading to product requirements just by answering direct questions based on the customer requirements taxonomy. Often, his answers led to requirements in other areas of the taxonomy as well. This occurred frequently in areas of the taxonomy with which he was less experienced. In these areas, his answers were less direct and often anecdotal in nature. While this still led to product requirements, it did not serve as a good test of the usefulness of the taxonomy in the elicitation process for areas of the corporate requirements taxonomy other than the highly technical area of manufacturing (his particular area of expertise). However, this example, coupled with a study done on the elicitation methodology for end user requirements (Hauge and Stauffer 1993), effectively covers the range of requirements through-

out the corporate requirements taxonomy, and shows the promise of the taxonomy as an information elicitation aid.

3.3. Representing Information

Once the requirements were elicited, the next step was to document the corporate requirements. This task presented two important challenges: (1) representing the requirements; and (2) organizing requirements for future use. These challenges highlighted the benefits of the taxonomy.

One important benefit of the corporate requirements taxonomy is that the same taxonomy is used for elicitation and storage. Requirements are elicited using the taxons as key words. The requirements are then stored under that same taxon. This makes elicitation easier, because the key words are provided. In addition, if a taxon has been covered it is filled with requirements, thus preventing skipped taxons. Using only one taxonomy structure also makes recalling requirements easier, because they are stored by the functional group and task that prompted them. In engineering information systems, it is rare that both questions and answers are filed in the same way. More commonly, information is elicited and then transposed to another system, where requirements are collated and manipulated.

3.4. Conceptual Design

Once the customer requirements (which include corporate requirements) were gathered and stored, conceptual design began. Our engineer began the conceptual design of the connector by using the

requirements (including both end-user and corporate) to develop initial concepts. Four initial concepts were developed which met the customer requirements. The product was split into two components; the casing and the contact. Then, two concepts were developed for each component. Combinations of the components result in four concepts (Fig. 5). The engineer then evaluated these concepts for functionality and manufacturability, and one concept was chosen to pursue further. This concept was then refined and re-evaluated.

Based upon the concepts' ability to fulfil pertinent requirements gathered during product definition, the groove casing and the straight contact were chosen as the concepts to pursue further. Based upon the negatives of the chosen concepts that arose during concept selection, an effort was made to improve the connector concept.

During conceptual design, our engineer used the corporate requirements taxonomy to extract requirements which were pertinent to the creation, evaluation and refinement of the concepts. The taxonomy helped to identify those requirements that were most important and most applicable to each component of the shorting connector. A slice of the corporate requirements taxonomy showing top level assembly and all of the requirements (with weights) contained therein is shown in Fig. 6. The engineer also had a computer-based version of the taxonomy which allowed recall by a sub-system, as in Fig. 7, which shows all customer requirements that are specific to the case. Information that is more specific was added to the requirements during conceptual design. The added information included trade offs, design attributes, and target values.

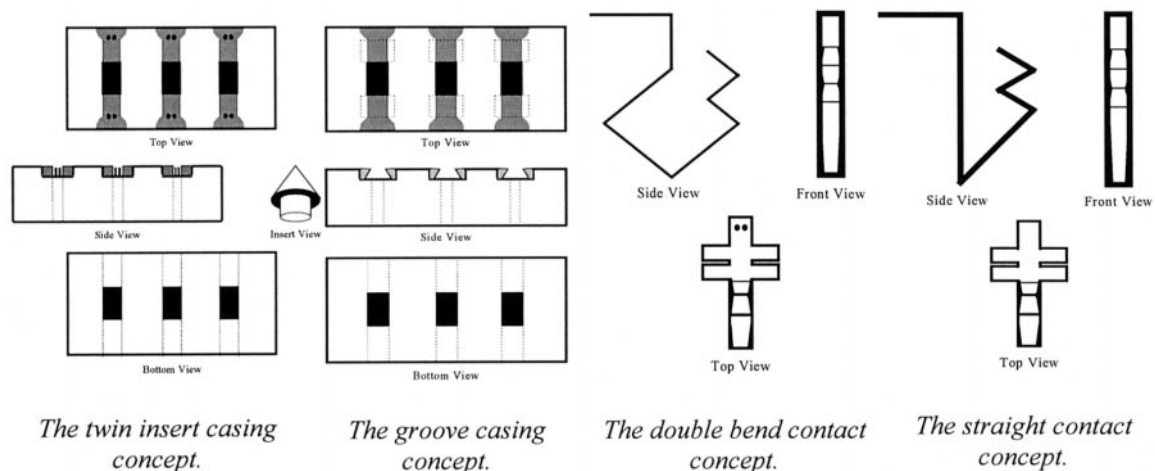


Fig. 5. The four concepts for the shorting connector.

Top Level Assembly minimize assembly operations-9, minimize time-9, minimize cost-3

1 Process Layout (Organizing the top level assembly process)

1.1 Rate (Number of products processed/unit time) **make assembly quick as possible-9**2 Staging (Where the different processes are performed) **able to do assembly at a table-9**3 Materials Handling (Handling of the parts during top level assembly) **minimize handling-9, minimize time-9**3.1 Type (Method of handling) **all assembly doable manually-9**3.1.1 *Geometry* **fits in hand-1, easily carried-1, minimize number of pieces-9, avoid tangling and nesting of parts-3**3.1.2 *Surface Condition* **contact manageable by hands without damaging the surface-3**

3.2 Ergonomics (Interaction between people and machinery during handling)

3.2.1 *Geometry* **minimize sharp corners-3, fits in hand-1**3.2.2 *Material* **casing stiff enough to squeeze-3**

3.3 Fragility (Product, part, and process robustness)

3.3.1 *Geometry* **minimize snags-3, minimize sharp corners-1**3.3.2 *Material* **not too bendable that contact can be bent in box or scratched-3**

3.4 Interfaces (Bins, containers, pallets, or anything else used in handling and interfacing with the product)

3.4.1 *Geometry* **must fit in palm of hand-3**4 Job Design (Assembly of the parts and product) **minimize time-9, minimize cost-9, minimize possibility of large scale mistakes in assembly process-9, minimize number of machines-3**

4.1 Manual doable by hand-3

4.1.1 Insertion (Putting one part onto or into another) **make quick and easy-9, minimize number of fasteners-3, minimize different types of fasteners-3**4.1.1.1 *Geometry* **can't put in backwards-3, put contacts through the plastic-3, can't put plastic upside down-3, won't snag going in-3, can't put in too far-3, minimize assembly directions-3, make components symmetrical or very asymmetrical-3, make hole big enough-3**4.1.1.2 *Features* **can't put in backwards-3, can't put in too far-3, feed back for far enough-3, can't put in upside down-3**4.1.1.3 *Surface Condition* **smooth plastic/contact slide-3**4.1.1.4 *Facilities* **minimize number of tools necessary-3**4.1.2 Orienting (Moving a part so it is facing the right way) **minimize the necessity-1**4.1.2.1 *Geometry* **one correct orientation-3, can insert any contact into any of the six holes-3**4.1.2.2 *Features* **make correct orientation obvious-3**4.1.3 Fixturing (Holding down a part or product for top level assembly) **doable with one hand-3**4.1.3.1 *Geometry* **make plastic big enough to hold in one hand-3, complete insertion while plastic rests on bench-1, flat bottom to plastic-3**4.1.4 Ergonomics (Interaction of people and machinery during top level assembly) **make process easy to perform-3**4.1.4.1 *Geometry* **fit fingers down to press in-3, hold wire end-3, minimize sharp corners-1**4.1.4.2 *Tolerances* **loose tolerances so can eyeball it in-3**4.1.4.3 *Material* **stiff casing to hold it by-3**

Fig. 6. The top level assembly portion of the corporate requirements taxonomy, with all pertinent shorting connector requirements and their relative weights.

As the conceptual design phase progressed, looking for applicable requirements became increasingly important. However, the meaning of the words used in the requirements became less clear. The requirement explanations, which are stored with the requirements, were used to clarify the meaning of the requirements (note: clicking on the requirements in Fig. 7 leads to additional information). Clarification was needed because, as the requirements elicitation session progressed, a short-hand developed between the engineer and the elicitor, and the requirements

became more succinct. This truncation of requirements had progressed to the point of unknowingly creating indecipherable requirements.

At this point in the exercise, the value of the taxonomy for retrieving information was demonstrated. The number of requirements that could be stored and accessed from the designer's memory was far fewer than those that had been stored in MOOSE. As the design process became iterative, requirements could be quickly referred to by searching through the taxonomy to find them. In addition, knowing the

Requirements List		
Customer Requirement	Taxonomy Level	Product
keep simple	2.1.1.4.9.1	Shorting Connector
casing stiff enough to squeeze	2.1.4.2.3.4	Subsystem
minimize snags	2.1.4.2.4.1	Casing
make plastic big enough to hold in one hand	2.1.4.3.1.4.1	
complete insertion while plastic rests on bench	2.1.4.3.1.4.1	
flat bottom to plastic	2.1.4.3.1.4.1	
fit fingers down to press in	2.1.4.3.1.5.1	
stiff casing to hold it by	2.1.4.3.1.5.4	
leave sides open for pulling out relay during maintenance	2.3.2.1.1	
minimize time to identify plastic for separation at recycling	2.4.4.3	
allow for plastic coding to aid separation	2.4.4.3.2	
leave blank section on plastic for coding to aid separation	2.4.4.3.2.1	

Fig. 7. Casing subsystem – specific requirements as saved in the MOOSE program for use in conceptual design of the shorting connector. Taxonomy level corresponds to the numbering system for organizing the taxonomy, as shown in Fig. 6.

taxon under which the requirement was stored enabled the designer to better understand the meaning of the taxonomy.

This case example demonstrates the prime advantage of the taxonomy. It defines a structure for the product definition information that can be intuitively understood by the design engineer. In this way, the leaf taxons supply clues as to the types of requirements needed. The designer can later go back to these requirements by accessing the same knowledge structure represented by this taxonomy. This process is analogous to storing and retrieving documents in a set of filing cabinets. Every document is located in a manila file folder in a certain hanging file folder in a certain drawer in a certain filing cabinet in the room. The taxonomy provides a system to label all of these components.

4. Summary

We have presented a taxonomy of corporate requirements for product design, and detailed the principles behind its creation. The taxonomy represents the needs of a corporation that may affect the form or function of a product. The taxonomy helps the gathering of requirements by detailing the issues that need to be covered. It eases the storage and retrieval of product design requirements by adding logic and structure to the process. The taxonomy eases the keeping of design histories by allowing for a single product requirements source that can be updated. This process was demonstrated during the design of an electrical switch with a local manufacturer. The taxonomy is useful for product definition, using QFD or similar methodologies.

References

- Anderson DM. (1990) Design for Manufacturability: Optimizing Cost, Quality, and Time-to-Market. CIM Press, CA
- Baya B, Leifer LJ. (1994) A study of the information handling behavior of designers during conceptual design. Proc 1994 ASME Design Technical Conf – 6th Int Conf on Design Theory and Methodology, Minneapolis, Minnesota
- Bossert JL. (1991) Quality Function Deployment – A Practitioner's Approach. ASQC Quality Press, Wisconsin
- Bralla JG. (1986) Handbook of Product Design for Manufacturing. McGraw-Hill, New York
- Chen A et al. (1990) Design history knowledge representation and its basic computer implementation. Proc 1990 ASME Design Technical Conf 2nd Int Conf Design Theory and Methodology, Chicago, IL
- Chen AT, Ward AC. (1991) Unifying QFD, Pugh Concept Selection, Function Structures, & Morphological Analysis. University of Michigan Department of Mechanical Engineering and Applied Mechanics Design Laboratory
- Coyle JJ, Bardi EJ, Cavinato JL. (1990) Transportation. West, New York
- Derr RL. (1973) A Taxonomy of Social Purposes of Public Schools. David McKay, New York
- Dieter GE. (1983) Engineering Design: A Materials and Processing Approach. McGraw-Hill, New York
- Dunn G, Everitt BS. (1982) An Introduction to Mathematical Taxonomy. Cambridge University Press, New York
- Fiksel J. (1991) Requirements management for concurrent engineering. Proc Concurrent Engineering Design Clinic, May 7–8, San Diego, CA
- Gershenson JA, Ishii K. (1991) Life-cycle serviceability design. Proc 1991 ASME Design Technical Conf 3rd Int Conf Design Theory and Methodology, Miami, FL
- Gershenson JA, Khadilkar D, Stauffer LA. (1994) Organizing and managing customer requirements during the product definition phase of design. Proc 1994 ASME Design Technical Conf 6th Int Conf Design Theory and Methodology, Minneapolis, Minnesota
- Gershenson JA, Stauffer LA. (1995) The creation of a taxonomy for manufacturability design requirements. Proc 1995 ASME Design Technical Conf – 7th Int Con Design Theory and Methodology, Boston, Massachusetts
- Gershenson JK, Stauffer LA. (1996) The validation of a taxonomy for manufacturability design requirements. Proc 1996 ASME Design Technical Con – 8th Int Conf Design Theory and Methodology, Irvine, CA
- Goldman A Slattery T. (1964) Maintainability: A Major Element of System Effectiveness. John Wiley & Sons, New York
- Hauge PL, Stauffer LA. (1993) Elk: a method for eliciting knowledge from customers. Proc 1993 ASME Design Technical Conf 5th Int Conf Design Theory and Methodology, Albuquerque, NM
- Hauser JR, Clausing D. (1988) The house of quality. Harvard Business Review
- Henstock EM. (1988) Design for Recyclability. Institute of Metals, New York
- Hundal MS. (1991) Systematic designing for cost. Proc 1991 ASME Design Technical Conf 3rd Int Conf Design Theory and Methodology, Miami, FL
- Jeffrey C. (1982) An Introduction to Plant Taxonomy. Cambridge University Press, New York

- Keys L.K. (1990) System life cycle engineering and DF'X'. IEEE Trans Components, Hybrids, and Manufacturing Technology 13(1):83-93
- Khadilkar D, Gershenson JA, Stauffer LA. (1994) Development of a database management tool for organizing conceptual phase design information. Proc 1994 ASME Design Technical Conf 14th Int Computers in Engineering Conf, Minneapolis, Minnesota
- King B. (1987) Listening to the voice of the customer: Using the quality function deployment system. National Productivity Rev. Summer:227-281
- Kusiak A. (ed) (1991) Concurrent Engineering: Theory and Practice. John Wiley & Sons, New York
- Navinchandra D. (1991) Design for environmentability. Proc 1991 ASME Design Technical Conf 3rd Int Conf Design Theory and Methodology, Miami, FL
- Ramaswamy R, Ulrich K. (1992) Augmenting the house of quality with engineering models. Proc 1992 ASME Design Technical Conf 18th Design Automation Conf, Scottsdale, AZ
- Slaughterbeck-Hyde R. (1989)i; A Measure of Design Quality. MS Thesis, University of Idaho
- Stauffer LA. (1988) An Empirical Study on the Process of Mechanical Design. PhD Disertation, Oregon State University
- Stauffer LA, Slaughterbeck-Hyde RA. (1989) The nature of constraints and their effect on quality and satisficing. Proc 1989 ASME Design Technical Conf 1st Int Conf Design Theory and Methodology, Montreal, Quebec, Canada
- Trucks H.E. (1987) Designing for Economical Production. SME Publications, MI
- Ullman DG. (1989) A taxonomy of mechanical design. Proc 1989 ASME Design Technical Conf 1st Int Conf Design Theory and Methodology, Montreal, Quebec, Canada
- Vonderembse MA, White GP. (1991) Operations Management: Concepts, Methods, and Strategies. West, New York
- Wheelan TL, Hunger DJ. (1989) Strategic Management and Business Policy. Addison-Wesley, New York
- Wright P, Pringle CD, Kroll MJ. (1992) Strategic Management: Text and Cases. Allyn and Bacon, MA